

Quotient Laws of ExponentsQuestion Bank - Additional 3.02 **ANSWER KEY***Show your work. Circle your final answer.***For questions 1 through 3, show the quotient of powers law, then evaluate.**

1. $\frac{5^{10}}{5^7}$

$$5^3 = 125$$

2. $\frac{7^5}{7^4}$

$$7^1 = 7$$

3. $\frac{(-4)^4}{(-4)^2}$

$$(-4)^2 = 16$$

Two negative factors: positive.

4. When raising a fraction to a power, the numerator and the denominator are each raised to the _____ power.

same**For questions 5 and 6, apply the power of a quotient law. Do not simplify the fraction.**

5. $\left(\frac{-3}{-6}\right)^6$

$$\frac{(-3)^6}{(-6)^6}$$

Both parts positive after an even power; the value is $(1/2)^6 = 1/64$.

6. $\left(\frac{7}{2}\right)^4$

$$\frac{7^4}{2^4}$$

$$= 2401/16.$$

The zero exponent.

7. Fill in the missing value in the pattern:
- $5^4 = 625$
- ,
- $5^3 = 125$
- ,
- $5^2 = 25$
- ,
- $5^1 = 5$
- ,
-
- $5^0 =$
- _____

1

8. Evaluate 12^0 and $\left(\frac{8}{9}\right)^0$.

1 and 1

For questions 9 and 10, rewrite the expression using a law of exponents, name the law, then evaluate.

9. $\frac{3.2^3}{3.2^1}$

$3.2^2 = 10.24$ — *quotient of powers*

10. $\left(\frac{1.5}{0.5}\right)^3$

$\frac{1.5^3}{0.5^3} = \frac{3.375}{0.125} = 27$ — *power of a quotient (or $3^3 = 27$)*

Working backwards.

11. Find the value of n that makes each statement true.

a. $\frac{2^n}{2^4} = 2^5$

$n = 9$

b. $\left(\frac{2}{3}\right)^n = \frac{16}{81}$

$n = 4$

c. $\frac{9^6}{9^n} = 1$

$n = 6$

12. Which expression is equivalent to $\frac{6^{10}}{6^5}$?

☐ A. 1^5

☐ B. 6^2

☐ C. 6^{15}

☒ D. 6^5

D

13. Select ALL of the expressions whose value is 1.

☒ **A.** $\frac{7^2}{49}$

☐ **B.** 0^5

☐ **C.** $\frac{2^5}{2^4}$

☒ **D.** $\left(\frac{3}{7}\right)^0$

☒ **E.** $\frac{2^5}{2^5}$

☒ **F.** 8^0

A, D, E and F

A: $7^2 = 49$, so $49/49 = 1$.